

CRYSTALLOGRAPHY

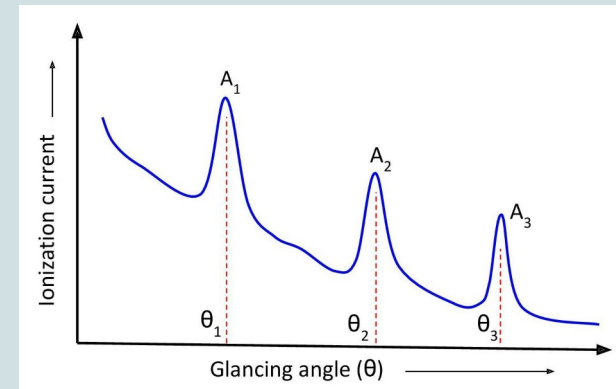
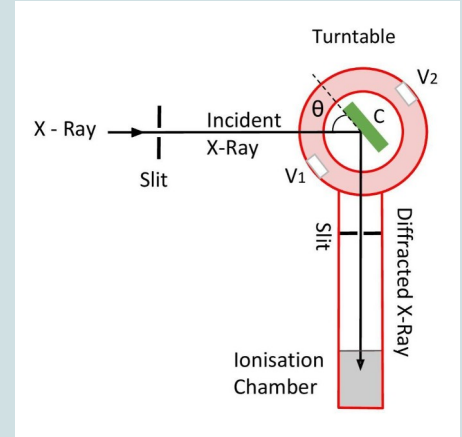
Bragg's X-ray Spectrometer

1. It is an apparatus to find the structure of single crystal using X-rays. It is used to measure *glancing angle θ* .
2. X-rays from X-ray tube is collimated using two adjustable slits S1 and S2.
3. The Vernier scale V is attached to table on which the crystal is placed.
4. A strong monochromatic X-ray beam is incident on the crystal face.

The reflected beam after passing through the slit S₃ enters the ionization chamber.

5. According to **Bragg's equation** $2d\sin\theta=n\lambda$
6. For $n = 1, 2, 3$ from the graph: $\sin \theta_1 : \sin \theta_2 : \sin \theta_3 = 1 : 2 : 3$
7. From observed values of θ and known values of d & n , the wavelength of the X-ray can be calculated.

9 times for 5 marks



Determination Of Crystal Structure

1. Set of planes (100), (110) AND (111) are obtained for angles θ_1 , θ_2 , θ_3 respectively in the first order reflection.
2. $2d_{100} \sin \theta_1 = \lambda$; $2d_{110} \sin \theta_2 = \lambda$; $2d_{111} \sin \theta_3 = \lambda$;
Hence, $d_{100} : d_{110} : d_{111} :: 1 / \sin \theta_1 : 1 / \sin \theta_2 : 1 / \sin \theta_3$
1. θ_1 , θ_2 , θ_3 can be obtained from graph Intensity vs angle i.e. where the peak occurs .
2. The reason for selection of planes (100), (110) and (111) is that these are the planes rich enough in terms of atoms.
3. The ratio of d_{100} , d_{110} and d_{111} for SC, BCC and FCC as follows,
SC :- $1 : 1/\sqrt{2} : 1/\sqrt{3}$
BCC :- $1 : 1/\sqrt{2} : 1/2\sqrt{3}$
FCC :- $1 : 1/2\sqrt{2} : 1/\sqrt{3}$
1. By comparing their ratio with the above equation, one can determine the crystal structure.

Miller Indices

2 times for 5 marks

- Miller Indices of a particular atomic plane are *reciprocals of intercepts of that plane on co-ordinate axis*.

